

DELAY-EMBEDDING ANALYSIS OF NEURAL NETWORK TRAINING LOGS: PRECONDITIONS, NEGATIVE RESULTS, AND VALIDATED DRIVER RECOVERY

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ABSTRACT

Empirical dynamic modelling promises to recover the structure of a high-dimensional system from scalar observations, which makes it attractive for monitoring neural network training from logs that consume negligible memory. This report establishes when that promise is redeemable. The embedding theorems of Takens and Stark reconstruct a compact invariant set that the trajectory revisits. We measure recurrence directly and find it absent over a whole run, then test the weaker hypothesis that short windows are locally stationary, logging every optimisation step to make that testable. The training loss does recur locally, but the binding constraint is the number of independent observations, which no logging rate changes: the correlation time is 163 steps against a transition of 1615, so the entire transition affords ten independent samples whether rows are written every step or every tenth. We then report three candidate signals for the delayed generalisation phenomenon known as grokking, each of which separates training runs convincingly and none of which survives a controlled comparison: an intrinsic-dimension estimate that reproduces closed-form constants of a straight line, a function-space velocity that tracks weight decay rather than generalisation, and a shared-manifold coupling that tracks configuration rather than the transition. Against these we set a positive result obtained in the regime where Stark’s theorem does apply. Convergent cross mapping recovers an externally injected driver from a single one-dimensional loss log in seven of eight coupled runs across two architectures, with no false positives among the control runs whose driver is logged but never applied, and with a limitation for strictly periodic drivers that we characterise quantitatively. Testing the reverse cross map, which the ground truth of these runs fixes by construction, shows that the dominant direction is recovered in seven of eight coupled runs while controls converge in neither direction, and also that unidirectionality is not demonstrable when the driver acts on the loss without delay. Finally, a factorial study separating the training split from the initialisation shows that the delay is itself a broadly distributed random variable, with thirty runs of one configuration spanning 1290 to 5600 steps and a single run of the same configuration reaching 11 790.

Keywords: empirical dynamic modelling, delay embedding, surrogate data, convergent cross mapping, grokking, training diagnostics.

1 INTRODUCTION

A training run exposes very little of itself cheaply. Weights and activations are large and expensive to store, whereas scalar logs such as the training loss, the validation loss and the parameter norm cost almost nothing. If the geometry of the underlying optimisation could be reconstructed from those scalars, monitoring would become inexpensive and architecture independent. This is the premise of

empirical dynamic modelling, and it rests on the embedding theorems of Takens (Takens, 1981) and, for forced systems, of Stark (Stark, 1999; Stark et al., 2003).

The premise is attractive for grokking, the delayed generalisation reported by Power et al. (2022), where a scalar precursor of the transition would be valuable and where the weight norm is an obvious candidate observable. We evaluate that candidate and two others, establish where delay embedding is and is not licensed on training logs, and report a positive result in the regime where it is.

The organising claim is the following. The embedding theorems reconstruct a compact invariant set that the trajectory revisits. A training run driven by an external signal with its own attractor satisfies this requirement by construction. The intrinsic optimisation transient does not. Results that respect this boundary reproduce across architectures and controls; results that cross it recover artefacts of trajectory smoothness or of run configuration. Section 2 states the requirements every claim in this report is held to. Section 3 then measures the precondition, section 4 reports three signals that violate it, section 5 reports the positive result obtained inside it, and section 6 measures how variable the transition itself is.

2 VALIDATION REQUIREMENTS

Several delay-embedding statistics return a finite number on data satisfying none of the hypotheses behind them, and three of the estimators used below were found during this work to be circular, mis-specified or underpowered in exactly that way. A result is therefore only as good as the controls reported with it. We fix the following requirements in advance and hold every claim in this report to them. Each costs well under an hour of computation.

1. **Measure the precondition before invoking a theorem.** Report recurrence first. If the trajectory does not revisit its past states, the language of attractors and their dimensions does not apply, whatever number an estimator returns. Section 3 does this and finds the precondition unmet on raw training logs.
2. **Calibrate every estimator on a system with a known answer, against a reference matched to the data in length and in sampling density.** Neither is a detail. A Lorenz attractor of 12 000 samples grows by a factor of 1.66 across the embedding range while one of 2000 samples grows by 8.86, so a reference of the wrong length manufactures a separation by itself (Fig. 3b). Density matters just as much: sampled at $dt = 0.01$ the same attractor fails a surrogate test of determinism at $p = 0.400$, and only at $dt = 0.1$ does it pass at $p = 0.010$. A negative result is uninterpretable unless a reference at comparable density is positive.
3. **Justify preprocessing per series rather than adopting it globally.** Subtracting a moving average manufactures structure in a sparse series, and differencing is a high-pass filter that removes exactly the slow drivers one may be looking for: the periodic drivers of section 5 cross map at 0.09 and 0.12 after differencing and at 0.81 and 0.80 without it.
4. **Test every claim against surrogates, with an ensemble large enough to express the threshold.** A statistic that does not exceed an amplitude-adjusted Fourier surrogate ensemble is reporting the linear autocorrelation of the series. Where a driver may be periodic, add a rotation null, and note that neither null has power in the strictly periodic case (section 5.3). With 39 surrogates the smallest attainable rank p-value is $1/40 = 0.025$, which coincides with the Bonferroni threshold for two nulls and leaves no room for the correction; we use 199.
5. **Include a control that differs from the positive runs in one respect only.** The ghost drivers of section 5 are the model: identical architecture, schedule, loss trajectory and logging, differing only in whether the coupling exists. Controls that differ in several respects at once cannot discriminate, which is how each signal of section 4 came to look convincing.
6. **For a causal claim, test the reverse direction.** A cross map computed in one direction establishes coupling, not causation. Run both directions at a matched embedding dimension, calibrate the convention on a system whose answer is known, and read direction from convergence rather than from skill, which section 5.4 shows reaches 0.57 in the false direction.

7. **Hold the configuration fixed, vary only seeds, and record the expected sign in advance.** A statistic proposed as a predictor of the transition must correlate with the delay across runs of one configuration. This test is inexpensive and it falsified the most promising of the candidates in section 4.

3 PRECONDITIONS FOR DELAY EMBEDDING

Takens’ theorem and its extensions guarantee that a delay map is an embedding of an *invariant set*. The guarantee is asymptotic and geometric: the orbit must return near its past states, so that neighbours in the reconstructed space are dynamically related rather than merely adjacent in time. When this fails, nearest-neighbour statistics measure the local tangent to the trajectory instead of the geometry of an attractor, a distortion first identified by Theiler (1986).

We therefore measure recurrence before invoking any theorem. For a delay-embedded series we fix a neighbourhood radius once, as a low quantile of all pairwise distances, and report the recurrence rate as a function of a temporal exclusion window. Fixing the radius once is essential; recomputing it after each exclusion would force the recurrence rate to equal the quantile for any input whatsoever, an error we made and corrected during this work. The shape of the resulting profile separates the two cases of interest. On an invariant set the rate falls to a positive plateau, because genuine returns occur at all temporal separations beyond the orbit’s period. On a transient the rate decays towards zero, because every close pair was adjacent in time.

The estimator is calibrated on systems with known answers. A Lorenz-63 trajectory retains 0.93 of its recurrence rate at the widest exclusion, whereas a monotone ramp, which has no dynamics at all, decays to exactly 0.00. Figure 1 applies it to the training logs.

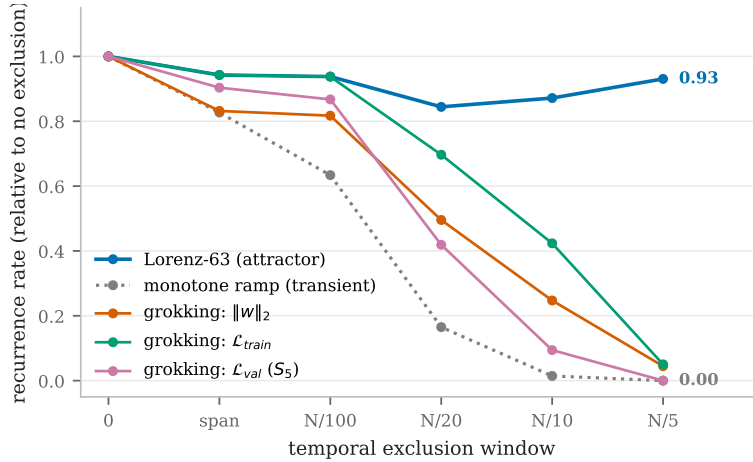


Figure 1: Recurrence rate as a function of the temporal exclusion window, normalised by the rate at zero exclusion. The Lorenz attractor holds a plateau, while all three training observables decay towards the monotone-ramp reference. Over a whole run the trajectories do not revisit their past states, so no single invariant set is available for the embedding theorems to reconstruct. Section 3.1 examines whether shorter windows escape this, and finds that the training loss does recur locally while the weight norm does not.

All three training observables behave like the transient reference rather than like the attractor. The validation loss of the S_5 runs reaches a ratio of 0.000 and the weight norm falls between 0.00 and 0.07.

That measurement is about the run as a whole, and by itself it settles less than it appears to. Failing to return to its past states over twenty thousand steps does not prevent a trajectory from being usefully stationary over a few hundred, and delay embedding applied inside such a window would be legitimate. The global measurement rules out treating a run as one invariant set; it does not rule out a local analysis. We therefore test the local hypothesis directly.

3.1 THE SAMPLING INTERVAL, AND WHAT IT CAN AND CANNOT FIX

A local analysis has two requirements that pull against each other. The window must be short enough that the dynamics do not drift appreciably across it, and it must contain enough independent observations to estimate anything. At one row per ten optimisation steps these cannot both be met, and the failure is severe: a 300-sample window, the length at which the dimension estimate of section 4.1 is computed, spans 3000 optimisation steps against a median memorisation-to-generalisation gap of 2875. The window is longer than the transition it is meant to resolve.

We therefore re-ran three conditions logging every optimisation step. Only the inexpensive series need this rate, since the training loss is already computed by the update and the parameter norm is a reduction over the weights, whereas a validation pass is a forward pass over the held-out split; the two are logged on separate schedules, and a test asserts that neither the logging rate nor a skipped evaluation perturbs training.

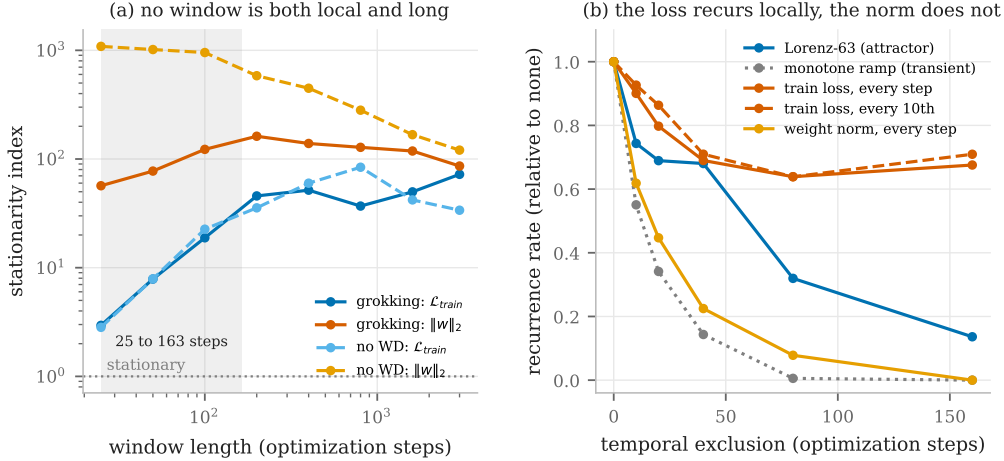


Figure 2: (a) Stationarity index against window length, in units of the value white noise of the same length produces, so that unity means indistinguishable from stationary. The normalisation is necessary: the unnormalised ratio falls as $1/\sqrt{L}$ for a stationary series, so a fixed threshold would report long windows as more stationary than short ones by arithmetic alone. The shaded band runs from the longest stationary window, 25 steps, to one correlation time of the training loss, 163 steps; a usable window would have to be wider than the band, and none is. (b) Recurrence inside an 800-step segment against temporal exclusion in optimisation steps. The training loss plateaus near the Lorenz reference while the weight norm decays to the monotone-ramp reference, and logging ten times more densely leaves both curves unchanged.

Three results follow, and the first corrects the impression left by Fig. 1. *Locally, the training loss does recur.* Inside an 800-step segment its recurrence profile plateaus between 0.64 and 0.69 against a Lorenz reference of 0.68, while the monotone ramp decays to 0.00 (Fig. 2b). The statement that training logs never revisit their past states holds globally and does not hold at local scale.

The weight norm does neither. It decays to 0.00 like the ramp, and its stationarity index never falls below 57 at any window length from 25 to 3000 steps, reaching 1087 for the run without weight decay. The observable on which the dimension estimate of section 4.1 is computed is therefore not rescued by working locally, which is a stronger statement than the global recurrence measurement supports.

Denser logging is not what was missing. The dense and decimated recurrence profiles agree throughout Fig. 2(b), so the earlier result was never an artefact of undersampling.

What does bind is the number of independent observations, and it is invariant to the logging rate. The count of independent samples in a window is the window length divided by the correlation time, and neither quantity depends on how often a row is written. Inside the transition the training loss has a correlation time of 163 steps against a gap of 1615:

logging interval	rows in the transition	independent samples
every ten steps	161	10
every step	1615	10

Ten. Logging ten times more densely multiplies the rows by ten and the independent samples by one. Against this, Fig. 3(b) puts the length needed to resolve even a Lorenz attractor at $n \gtrsim 6000$, a shortfall of roughly six hundred times. The two constraints close on each other from both sides: the longest window that is indistinguishable from stationary is 25 steps, which is shorter than a single correlation time, so there is no scale at which the series is simultaneously stationary and observed more than once independently.

The autocorrelation agrees with that reading. The dense training loss falls to 0.76 at lag one and is then flat near 0.72, which is the signature of an independent minibatch-sampling term of about a quarter of the variance riding on a slowly varying component, rather than dynamics that were hidden below the previous resolution. A Lorenz trajectory sampled at a comparable density instead decays smoothly through 0.87, 0.61, 0.38 and 0.23. Consistent with this, no training observable exceeds an IAAFT surrogate ensemble under simplex projection: the grokking training loss scores 0.739 against a surrogate mean of 0.722 at $p = 0.52$.

That last comparison requires a reference at comparable sampling density before it means anything, which is the requirement of section 2. A Lorenz trajectory sampled at $dt = 0.01$ is oversampled in the same way and also fails to beat its surrogates, at $p = 0.400$, despite being deterministic by construction; sampled at $dt = 0.1$ it passes at $p = 0.010$ with skill falling from 1.00 to -0.23 across the forecast horizon. We report both, and read the failure of the training logs only against the reference that passes.

One caveat constrains recurrence as a diagnostic. Independent noise also produces close pairs at arbitrary temporal separations and therefore also scores near unity. Recurrence separates smooth transients from everything else; it does not separate an attractor from noise, which is why the determinism test above is required alongside it.

4 SIGNALS THAT DO NOT SURVIVE CONTROLLED COMPARISON

4.1 INTRINSIC DIMENSION OF THE WEIGHT-NORM TRAJECTORY

The Levina–Bickel maximum likelihood estimator (Levina & Bickel, 2004) applied to a sliding window of the weight-norm series produces a curve that rises during memorisation and falls before generalisation. Four observations remove its interpretation as a dimension. The first is analytic; the last, established in section 3.1, is that the weight norm is never locally stationary at any window length tested, with a stationarity index between 57 and 1087, so no window of it can be read as a stationary system.

Proposition 1 (The estimate is a constant of a straight line). *Let the delay vectors be sampled at uniform intervals from a curve that is locally straight at the scale of the neighbourhood, and let a Theiler exclusion of W samples be applied. The k nearest neighbours of an interior delay vector are then its temporal neighbours at offsets $\pm(W+1), \pm(W+2), \dots$, so the ordered neighbour distances satisfy $r_j \propto s_j$, where s is the non-decreasing sequence $W+1, W+1, W+2, W+2, \dots$ truncated to k terms. The Levina–Bickel estimate*

$$\hat{d} = \left[\frac{1}{k-1} \sum_{j=1}^{k-1} \log(r_k/r_j) \right]^{-1} = (k-1) / \sum_{j=1}^{k-1} \log(s_k/s_j) \quad (1)$$

is therefore a function of k and W alone, and contains no information about the data. For $k = 5$ it equals 1.330 when $W = 0$ and 10.765 when $W = 14$.

The proposition is verified against the estimator in Fig. 3(a). Applied to a synthetic straight line, which possesses no dynamics whatsoever, the estimator returns the predicted constant to within 1.1% at every k and W tested. Applied to the two control runs trained without weight decay, which never generalise, it returns the same constants: medians of 1.33 and 1.38 without exclusion against a prediction of 1.33, and 11.30 and 11.69 with exclusion against a prediction of 10.76. Both plateaus are therefore reproduced by a straight line, and neither can be evidence about an attractor.

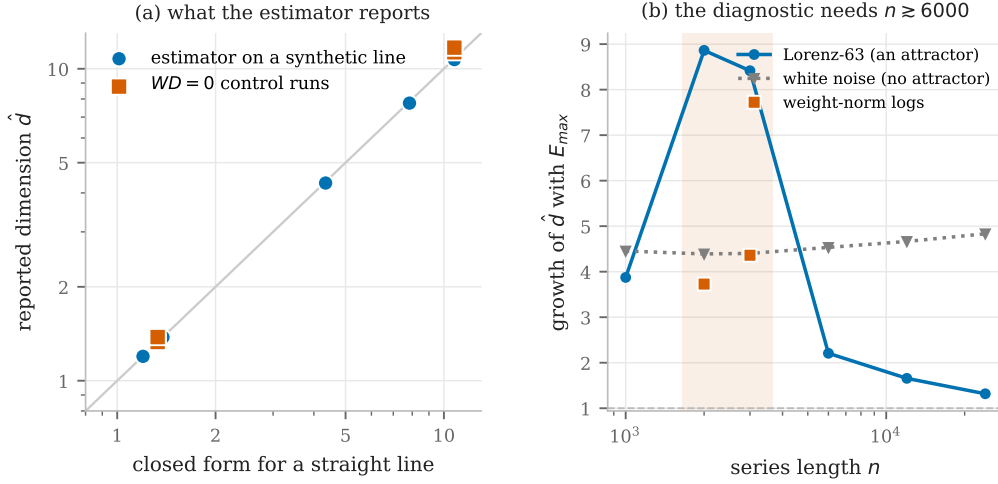


Figure 3: (a) Dimension reported by the estimator against the closed form of Proposition 1, on logarithmic axes with the identity line shown. A synthetic straight line and the $WD = 0$ control runs both lie on the prediction. (b) Growth of the estimate between $E_{max} = 5$ and $E_{max} = 30$, against series length. A quantity that is a property of the data does not depend on the space it is measured in, so a value near unity is the signature of a resolvable attractor. The Lorenz reference reaches it only beyond $n \approx 6000$; inside the shaded band, which is the length training logs provide, it scores worse than both white noise and the logs, so the diagnostic cannot discriminate there.

The second observation concerns identifiability. An intrinsic dimension exists only if the estimate is insensitive to the dimension of the space in which it is measured, so the growth of $\hat{d}$ between $E_{max} = 5$ and $E_{max} = 30$ is a natural diagnostic. It must, however, be read against a reference of the same length, because it is strongly length-dependent. Figure 3(b) measures that dependence. A Lorenz reference falls from a growth factor of 8.86 at $n = 2000$ to 2.21 at $n = 6000$ and 1.32 at $n = 24000$, while white noise stays between 4.4 and 4.8 throughout.

The consequence is that at the lengths training logs provide the diagnostic has no power. The two weight-norm series, of 2000 and 3000 samples, grow by factors of 3.73 and 4.36; a Lorenz attractor of the same length grows by 8.86, which is worse than both, and worse than noise. We therefore do not conclude that these logs resemble noise more than they resemble an attractor, since the measurement cannot support a statement in either direction. What follows instead is a statement about the data budget: at $n \leq 3000$ the estimator fails to resolve a textbook attractor, so no intrinsic dimension is identifiable from a log of that length whether or not one exists. This is a precondition on sample size, and section 2 states it as such. The case against the dimension signal does not rest on this diagnostic; it rests on Proposition 1 and on the seed comparison below.

Both observations so far assume a Theiler exclusion, and the estimate is often computed without one. That case deserves answering on its own terms, because the exclusion is not what removes the result. Proposition 1 evaluated at $W = 0$ returns 1.330, so switching the exclusion off does not restore information; it pins the estimate to the tangent constant, which is exactly why such curves look stable. The $WD = 0$ controls sit at that constant for their entire duration without ever generalising.

That argument disposes of the reported *level* but not of the reported *shape*. A control whose weight norm never departs from a straight line is flat by construction and cannot refute a claim about a rise followed by a fall. The control that can do so holds weight decay fixed at 1.0 and reduces the training fraction, so the same regulariser drives a non-stationary norm in a model that never generalises. Figure 4 reports that comparison at $W = 0$, $k = 5$, $E_{max} = 15$ and a window of 300 samples, that is, with no temporal exclusion at all.

The signal fails an internal consistency check before it reaches any control. Across three seeds of the identical grokking configuration, whose weight norms terminate at 29.89, 29.12 and 30.39 and so agree to within 4%, the estimate falls in one run (Spearman trend -0.54) and rises in the other two ($+0.94$ and $+0.90$). The descent is therefore not reproducible across seeds of the very condition that

produces it. The late-training level of those three runs is 1.51, 9.21 and 11.64, a spread of a factor of 7.7 within one configuration, which no dimension of a shared attractor can exhibit. Against the matched controls the direction is likewise uninformative: the never-generalising runs at the same weight decay give trends of $+0.75$ and $+0.22$, which lie inside the range spanned by the runs that do generalise.

What the estimate does track is the local smoothness of the series. Pooled over 1026 windows and all six runs, it attains a Spearman correlation of $+0.887$ with the ratio $\text{std}(\Delta x)/\text{std}(x)$, a statistic computed without any embedding, neighbour search or likelihood (Fig. 4c). Within individual runs the correlation ranges from $+0.32$ to $+0.996$, and the mapping is not identical across runs, so smoothness is a strong correlate rather than a complete description. The direction of the norm’s drift is immaterial to it, as expected: the grokking norm falls from 42 to 30 and the lowdata norm rises from 42 to 61, and a monotone rise is as locally straight as a monotone fall.

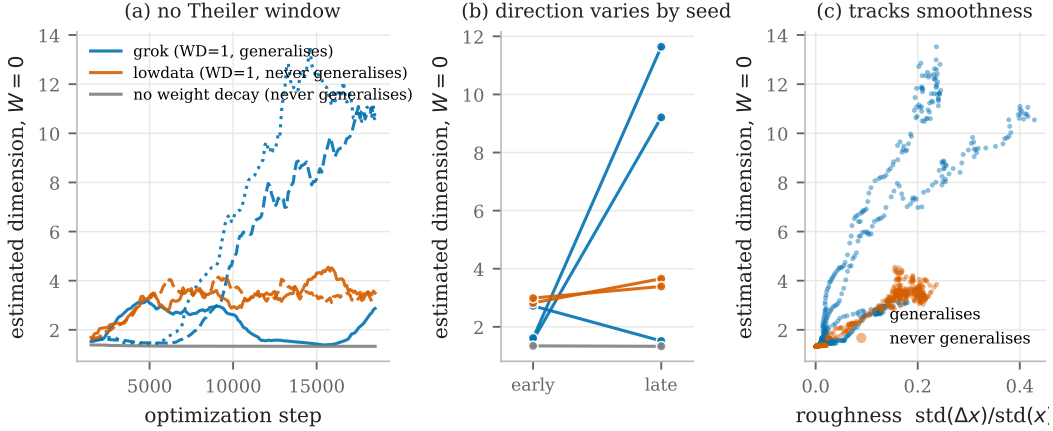


Figure 4: The dimension signal evaluated with no Theiler exclusion. (a) Sliding-window estimates. Line style distinguishes seeds within a family. (b) Median estimate over the first and last third of training. Two of the three generalising runs rise, one falls, and the never-generalising runs at identical weight decay lie between them; the no-weight-decay run is flat at the tangent constant, its apparent monotone decline spanning only 1.34 to 1.33. (c) The estimate against a roughness ratio that involves no embedding.

The third observation is procedural. The windows were labelled by their centres, so each estimate incorporated data recorded after the step it was assigned to. Under causal labelling, in which an estimate is assigned to the last step it observed, the reported lead over the transition does not survive.

Averaging the per-point estimates also inflates them relative to pooling the likelihood, as MacKay & Ghahramani (2005) note. The correction changes the level but not the conclusion, because the underlying quantity is not identifiable in the first place.

4.2 FUNCTION-SPACE VELOCITY

The preceding failure suggests an observable that cannot be a smoothness artefact of the weight norm. We therefore recorded, at each logged step, the logits of a fixed probe set of training inputs, centred each example’s logit vector over classes and normalised it to unit length. The resulting series is invariant to the uniform rescaling of logits that weight decay induces, so a change in it indicates that the computed function has moved rather than that its confidence scale has changed. The median per-example movement of the normalised logits separates a grokking run from a run without weight decay by a factor of approximately 360.

The separation does not survive its control. We trained runs in which weight decay is held at 1.0 and only the quantity of training data is reduced, so that the model memorises and never generalises. Figure 5 shows the outcome. The runs that never generalise sustain a *higher* velocity than the runs that grok. The comparison carries three seeds of the grokking condition against four never-generalising runs, at two training fractions with the smaller replicated across three seeds, and the two families do

not overlap. The only run far below the others is the one without weight decay, in which the function freezes within a few thousand steps and thereafter only rescales.

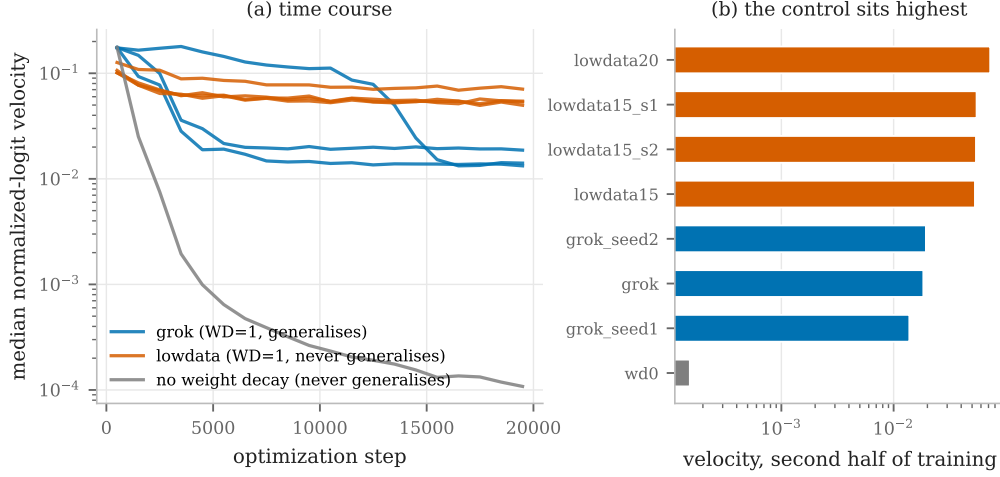


Figure 5: (a) Median velocity of the normalised probe logits against the optimization step, on a logarithmic scale. (b) The same statistic over the second half of training, one bar per run. If the velocity indicated impending generalisation, the runs that never generalise would sit low. They sit highest. Weight decay is identical in the blue and orange families, which differ only in the quantity of training data.

Quantitatively, the runs that never generalise span 5.30×10^{-2} to 7.26×10^{-2} over the second half of training against 1.38×10^{-2} to 1.94×10^{-2} for the grokking runs, a separation of a factor of 2.73 between the closest members of the two families. The three seeds of the reduced-data condition agree to within 3.4%, so the ordering does not rest on a single draw. The run without weight decay falls to 1.53×10^{-4} . Every run with weight decay lies within a factor of five of every other, and the factor of 360 quoted above is entirely accounted for by the presence or absence of regularisation. The statistic therefore measures the reorganisation that weight decay imposes on the function, and among regularised runs it is, if anything, inversely related to generalisation.

4.3 SHARED-MANIFOLD COUPLING

Both preceding failures are statistics of a single transient series. Cross mapping asks a different question, namely whether two observables lie on a common manifold, and that question is well posed on a transient because it requires no recurrence of either series alone. We therefore tested whether the training loss and the parameter norm, both available without any validation data, share a manifold, and whether the coupling changes at the transition.

The result appeared decisive. Figure 6(a) shows a clean separation with no overlap: the three runs with a delayed transition have median coupling between 1.00 and 1.15, while the four runs without lie between 2.87 and 5.61.

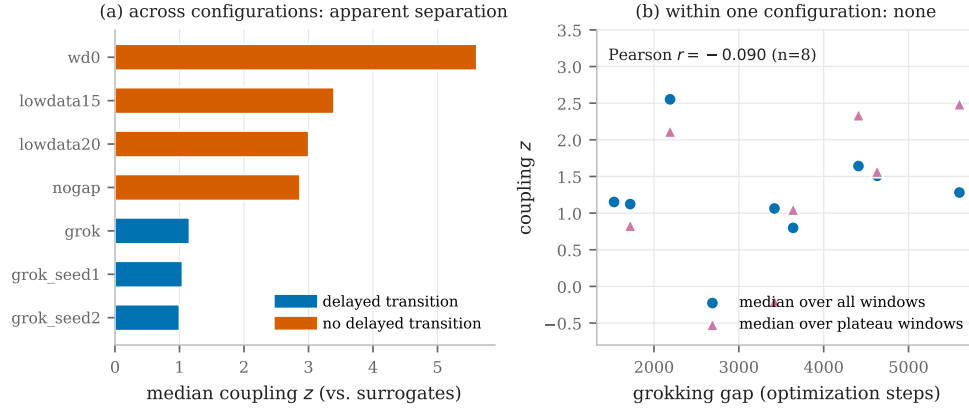


Figure 6: (a) Median coupling between the training loss and the parameter norm separates runs with a delayed transition from runs without. (b) The same statistic across eight runs of a single configuration, in which only the split and initialisation seeds vary. The gap spans 1290 to 5600 steps and the coupling is unrelated to it. The separation in panel (a) reflects configuration, not the transition.

Two checks dissolve it. Within individual runs the statistic does not move at the moment of generalisation; the apparent step in an aggregate over runs arises from averaging runs whose transitions occur at different steps. More importantly, the grouping in panel (a) coincides exactly with configuration, since the low group consists of the runs with a training fraction of 0.3 and weight decay of 1.0 and the high group contains every other setting. We therefore held configuration fixed and varied only the split and initialisation seeds, which the study of section 6 shows produce gaps between 1290 and 5600 steps. The expected sign was recorded before the runs were executed. The observed correlation between the gap and the coupling is -0.090 (Pearson) and $+0.214$ (Spearman) over eight runs, that is, nothing, and for plateau windows it is weakly positive, which is the opposite of the prediction. Within one configuration the coupling spans 0.80 to 2.55, against 1.0 to 5.6 across configurations.

4.4 A COMMON FAILURE MODE

The three signals fail in the same way, as table 1 summarises. Each separates *configurations* rather than *transitions*, and each is supported by a comparison in which the runs differ in more than the property of interest. The controls that falsify them share every feature with the positive runs except the mechanism under test: weight decay held fixed while data is varied, or configuration held fixed while seeds are varied. We regard this, rather than any individual negative result, as the principal methodological lesson.

Table 1: The three signals, the comparison that made each convincing, and the control that falsifies it. In every case the falsifying control differs from the positive runs in one respect only.

Signal	Apparent evidence	Falsifying control	What it measures
Intrinsic dimension of $\ w\ _2$	Rises then collapses before generalisation	A straight line reproduces both plateaus; across seeds of one configuration the estimate rises twice and falls once	Local smoothness of the series
Function-space velocity	Separates $WD = 1$ from $WD = 0$ by a factor of 360	$WD = 1$ with reduced data never generalises, yet sustains a higher velocity	Reorganisation imposed by weight decay
Shared-manifold coupling	Separates runs with a delayed transition from runs without, with no overlap	Across eight runs of one configuration the coupling is unrelated to the delay, $r = -0.090$	Run configuration

5 DRIVER RECOVERY IN THE REGIME WHERE THE THEOREMS APPLY

Stark’s extension of Takens’ theorem covers a system driven by an external signal that possesses its own attractor (Stark, 1999; Stark et al., 2003). A training run modulated by an injected driver satisfies that hypothesis by construction, which makes it the natural setting in which to ask whether anything at all is recoverable from a one-dimensional log.

5.1 DESIGN

We use two independent settings. The first consists of existing logs from a ResNet-18 trained on CIFAR-10, in which a fraction of each batch’s labels is corrupted according to a driver logged alongside the loss. The second reproduces the design in this work, using a one-layer transformer on modular addition with drivers under our control. In both cases the ground truth is known, and in both cases the essential control is a *ghost* driver, which is logged exactly like a real one while the batch is returned unmodified. A ghost run shares the architecture, the schedule, the loss trajectory and the logging cadence of a coupled run and differs only in whether coupling exists.

Detection uses convergent cross mapping (Sugihara et al., 2012): the response is delay embedded and used to predict the driver, on the reasoning that if the driver forces the response then the reconstructed manifold of the response carries information about the driver’s state. Significance is assessed against two nulls. The first uses iterative amplitude-adjusted Fourier surrogates (Theiler et al., 1992; Schreiber & Schmitz, 1996), which preserve the power spectrum and the exact value distribution of the driver while destroying its alignment with the response. The second rotates the driver in time, which preserves every value and changes only the alignment. No result is reported without them.

5.2 RESULTS

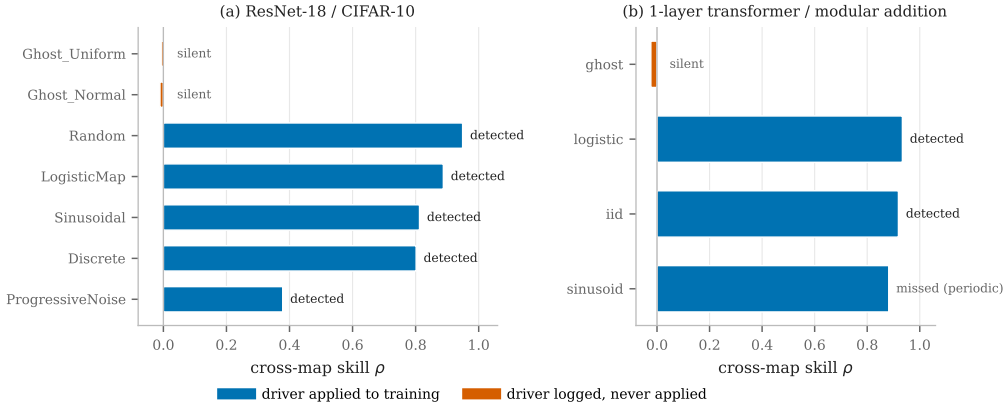


Figure 7: Cross-map skill for every run in both settings. Drivers that were applied to training are recovered from the loss log alone; drivers that were logged but never applied are not. The single missed detection is the smooth periodic driver discussed in section 5.3.

Figure 7 summarises both settings. In the ResNet setting all five coupled drivers are detected, with skills between 0.379 and 0.949 and $p \leq 0.05$ under both nulls, and all ghost runs remain silent at skills of -0.011 and -0.007 . In the transformer setting the chaotic and independent drivers are detected at 0.934 and 0.919 and the ghost remains silent at -0.024 . Across both settings, seven of eight coupled runs are detected and no ghost run produces a false positive. Three of the five ghosts carry a varying driver and cross map at -0.011 , -0.007 and -0.024 ; the remaining two carry a constant driver, which has no variation to recover and therefore cannot be cross mapped at all. We count them as controls but they are not evidence, and they are omitted from Fig. 7 for that reason.

The ghost runs carry the argument. They share the training trend with the coupled runs, so their near-zero skill excludes the explanation that defeated the signals of section 4, namely that a statistic separates runs because their loss curves have different shapes.

The detection rule declares a driver recovered when either null rejects, which is a family of two tests and therefore admits a false-positive rate approaching 0.10 rather than the nominal 0.05. The ensemble of 39 surrogates used above cannot express the correction, since its smallest attainable rank p-value, $1/40 = 0.025$, coincides exactly with the Bonferroni threshold for two tests. We therefore repeated both experiments with 199 surrogates, where the attainable floor is 0.005. The verdicts are unchanged: seven true positives, one false negative and no false positives under either the original rule or the corrected one. The single case that had been marginal, the ResNet periodic driver at $p = 0.050$, resolves to $p = 0.015$ with the larger ensemble and survives the correction. The counts reported here are therefore not an artefact of the multiplicity.

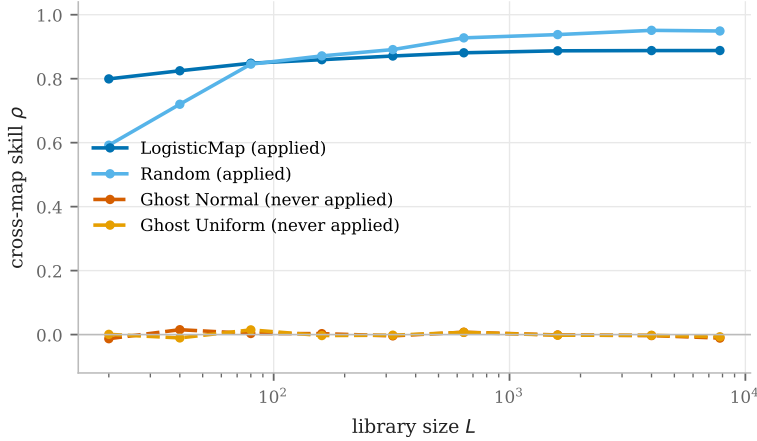


Figure 8: Cross-map skill against library size. Coupled runs improve as the manifold is sampled more densely, which is the convergence signature that distinguishes causation from coincidental correlation. Ghost runs remain at zero for every library size.

Figure 8 shows the convergence criterion that gives the method its name. The skill of the coupled runs rises with library size, most clearly for the independent driver, which climbs from 0.59 to 0.95. The ghost runs show no such rise, and no level.

One preprocessing choice deserves emphasis because it initially inverted the conclusion. Differencing the response is the obvious way to remove the training trend, but it is a high-pass filter and therefore removes slow drivers. The periodic drivers have the strongest direct coupling of any run, with correlations of 0.69 and 0.58 against the loss, yet they cross map at 0.09 and 0.12 after differencing and at 0.81 and 0.80 without it. The trend requires no manual removal, because both nulls and the ghost runs already control for it.

5.3 A LIMITATION FOR PERIODIC DRIVERS

The one missed detection is instructive and is not repaired by a better statistic. The smooth sinusoidal driver of the transformer setting yields a cross-map skill of 0.882, which is high, but its surrogates reach between 0.87 and 0.94, which is equally high. The reason is structural. The loss reveals the phase of the cycle, and any series of the same period, whether a spectrum-preserving surrogate or the driver rotated in time, is a deterministic function of that phase and is therefore equally predictable from it. Spectrum-preserving nulls consequently have no power against a strictly periodic driver. The corresponding driver in the ResNet setting is detected only because it is quantised to ninety-five distinct values, so its surrogates depart from it considerably further. A different null, for example a driver of a different period, would be required.

5.4 DIRECTION OF THE RECOVERED COUPLING

Detection is not direction. A statistic that recovers the driver from the loss establishes that the two series are coupled, and the causal reading requires in addition that the coupling be shown to run from the driver to the loss rather than the reverse. Cross mapping is asymmetric by construction and can be

asked this question directly. Embedding Y and predicting X , written $Y \text{ xmap } X$, is evidence that X drives Y (Sugihara et al., 2012), so the forward map is loss xmap driver and the reverse is driver xmap loss.

These logs are an unusually clean setting for the test, because the ground truth is known by construction rather than assumed. The driver schedule is evaluated from a fixed seed before any gradient is taken, so training cannot influence it and the coupling is unidirectional by design.

The convention itself is calibrated on a system with a known answer, since inverting it would invert every causal statement that depends on it. On coupled logistic maps (Sugihara et al., 2012) with $\beta_{xy} = 0$ and $\beta_{yx} = 0.32$, so that X drives Y alone, the true direction rises from 0.331 to 0.993 with library size while the false direction runs from 0.585 to 0.569 and does not converge. The false direction therefore reaches a skill of 0.57 while carrying no causal information at all, which establishes that convergence rather than skill is the discriminating quantity.

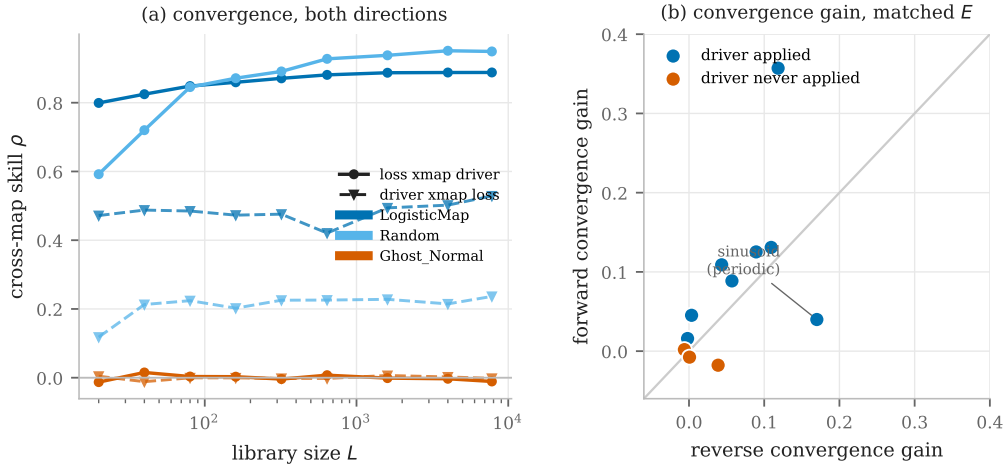


Figure 9: Forward against reverse cross mapping, both directions at a matched embedding dimension. (a) Skill against library size. The forward map converges for the coupled runs while the reverse remains flat, and both directions are flat for the ghost. (b) Convergence gain in the two directions. Coupled runs lie above the diagonal, ghost runs sit at the origin, and the exception is the smooth periodic driver of section 5.3.

Figure 9 reports the comparison. At a matched embedding dimension the forward map converges more strongly than the reverse in seven of the eight coupled runs, most clearly for the independent driver at gains of 0.357 against 0.118. The ghost runs converge in neither direction, with all gains within 0.039 of zero, so the test does not manufacture a direction where no coupling exists. The single exception is the smooth periodic driver of the transformer setting, whose reverse gain of 0.170 exceeds its forward gain of 0.040; this is the same run whose forward detection fails its null at $p = 0.77$, and it is excluded by section 5.3 on independent grounds.

The result should not be overstated, and a stricter reading of the same data does not support unidirectionality. If the reverse direction is granted the best of five embedding dimensions rather than being held at the forward's $E = 3$, it converges and beats its own surrogate null in most runs. Under a criterion that requires the reverse map to be silent, none of the eight coupled runs is assigned a unidirectional driver-to-loss reading and five are called bidirectional. Selecting the maximum over five embeddings is itself a multiple comparison and inflates the reverse, but the effect is not solely an artefact of that selection.

The explanation is structural rather than a defect of the estimator. The driver is applied to training at every step, so the loss depends on the driver contemporaneously, and a delay vector of the driver already contains the value that the loss is partly a function of. The reverse map can therefore succeed by direct functional dependence without reconstructing any manifold. Cross mapping does not separate causation from strong instantaneous coupling, and under strong forcing the response

becomes a function of the driver, which is the generalised-synchrony regime (Rulkov et al., 1995) known to compromise the method (Ye et al., 2015).

We therefore claim what the data support. The coupling is detected, and its dominant direction is recovered correctly in seven of eight runs, with the exception falling in the periodic case already excluded. Unidirectionality is not established, and demonstrating it would require a design in which the driver acts on the loss with a delay, so that the contemporaneous term is absent.

5.5 THE METHOD IS NOT DISCRIMINATIVE BETWEEN A RUN’S OWN METRICS

For completeness we applied the same procedure to pairs of observables within single uninjected runs. Eleven of sixteen directed pairs pass both nulls, with skills as high as 1.000. This is consistent with the theory, since the training loss, the validation loss and the parameter norm are three observations of one system, but it means that cross mapping between a run’s own metrics carries no information about the transition and should not be interpreted as a discovery.

6 VARIABILITY OF THE DELAYED TRANSITION

Any statistic proposed as a precursor of grokking is a statement about the delay, so the distribution of the delay is worth measuring before precursors are sought. The configuration seeds a single random stream from which the train and validation split is drawn and from which the model initialisation then continues. A change of seed therefore alters both which examples are trained on and what the initial weights are, and the two cannot be attributed separately. We separate them by restarting the stream between the two steps, verify that the modification is inert when unused and that it decouples the two factors in both directions, and run a factorial study of six splits by five initialisations with weight decay fixed throughout.

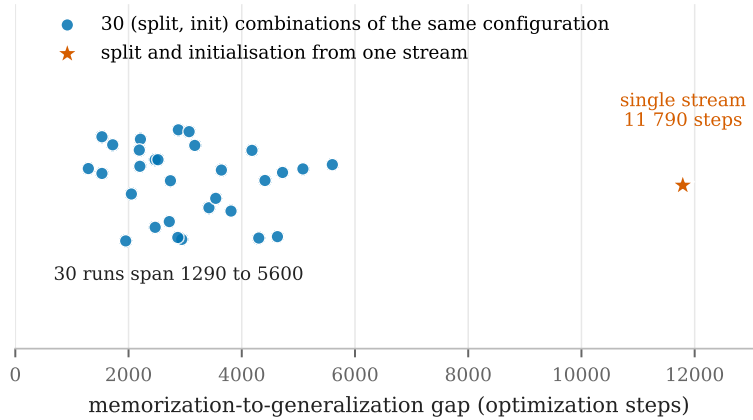


Figure 10: Memorisation-to-generalisation gap for thirty runs of one configuration, varying only the split and the initialisation seed. Every run generalises and every gap lies between 1290 and 5600 steps. Marked separately is a run of the same configuration in which the split and the initialisation are drawn from a single stream, whose gap of 11 790 steps lies far outside the distribution.

All thirty runs generalise, with gaps between 1290 and 5600 steps, a mean of 3062, a median of 2875 and a standard deviation of 1130. The delay is thus a broadly distributed quantity whose standard deviation is more than a third of its mean, within a configuration held otherwise fixed.

A run of the same configuration under the original coupled stream produced a gap of 11 790 steps, which is 7.7 standard deviations above the mean of the grid and 2.11 times its largest member, while two further runs at other seeds gave 1510 and 1570, both inside it. Five cells of the grid share the split of the outlying run and give gaps between 2050 and 4300, so the split alone does not account for it. The grid contains no cell reproducing the particular pairing of split and initialisation that the coupled stream produces, and we therefore attribute the outlier to that pairing rather than to the configuration. The practical consequence is that a delay measured from one run is not a property of

the configuration, and a precursor validated against one run has been validated against a single draw from a wide distribution.

The variance decomposition is inconclusive by design rather than by accident. The split accounts for 19.8% of the variance in the gap and the initialisation for 23.8%, with the remainder in the interaction. Neither factor dominates. A partial analysis of the first fifteen cells indicated the opposite, with the split accounting for 43% against 17% for the initialisation, which is a caution against reading factorial studies before they complete.

7 CONCLUSION

Delay-embedding methods are applicable to neural network training logs in a narrower regime than their premise suggests, and within that regime they work well. When an external driver possessing its own attractor forces training, convergent cross mapping recovers it from a single scalar loss log, in seven of eight coupled runs across two architectures and with no false positives among the runs whose driver was logged but never applied. The coupling is directional in the sense that the forward map converges more strongly than the reverse in seven of eight runs while the controls converge in neither, though these data do not establish unidirectionality, for the reason set out in section 5.4. The quantity requires no validation data and is obtained from a log that costs nothing to store.

The intrinsic optimisation transient does not admit the same treatment, and the reason is quantitative rather than a matter of taste. Its correlation time is a tenth of the transition it would have to resolve, so the transition affords about ten independent observations at any logging rate, against the thousands an attractor reconstruction requires. The three statistics we examined that appeared to detect the grokking transition were found instead to separate run configurations. The phenomenon itself is also more variable than a single run can reveal: thirty runs of one configuration produce delays between 1290 and 5600 steps, while a run of that same configuration with the split and initialisation drawn from one stream reaches 11 790.

One question is left open. Whether any training-side observable distinguishes a run that will generalise from a run that will not, with configuration held fixed, remains undecided: none of the five examined here, comprising the parameter norm, the trajectory roughness, the intrinsic dimension, the function-space velocity and the shared-manifold coupling, is able to do so. The driven regime is meanwhile the one in which the theory applies and validation against a known answer is possible, and it is where we would place further effort.

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